VIRAAT CHAUHAN • FULL-STACK SOFTWARE PROJECT • LOCAL-FIRST AI TRADING RESEARCH

Quant Labs

A local-first AI trading research system that turns scattered trades, notes, screenshots, scripts, and PDFs into a living memory graph.

The system connects evidence to trades and strategies so traders can ask why decisions worked or failed. Viraat Chauhan • Trading Intelligence OS • Paper-only MVP

Problem

Trading evidence is scattered across executions, screenshots, notes, PDFs, scripts, and journal entries. Current tools track trades, but they rarely explain **why** a decision worked or failed.

- Patterns disappear after the trading day.
- Rules live separately from the trades they affected.
- Review questions need evidence, not memory.

Approach

Quant Labs captures evidence, generates metadata, connects it to trades and strategies, and keeps the result searchable inside a local-first workspace.

Capture → Enrich → Connect → Review

Links and uploads become structured research records with tags, setup notes, entry rules, exit rules, and risk context.

Example Review Questions

"Why did my VWAP pullback fail after a loss?"

"Which sources support or contradict this ORB rule?"

Vault Capture

Auto Capture

Link

https://... | Import

Upload a file

PDF, image, CSV, Markdown, text, JSON, Pine, Python

Waiting for a link or file.

AUTO-FILLED FIELDS

Title	file/link metadata
Type	MIME / extension
Body	extracted text
Tags	generated + source
Strategy	setup + rules

Learning Vault

2 records

Search

Opening Range Breakout Playbook Article AI tags

Opening range breakout plan requiring relative volume confirmation, a clean premarket range, stop placement before entry, and no entries after 10:45 AM. The note highlights execution risks around chasing, early stop movement, and low-liquidity breakouts.

Strategy research · orb · risk-reward · opening-range

GENERATED TAGS

strategy:orb · entry:range-break · risk:predefined-stop · filter:rvol

STRATEGY INFO

Paper-test a stricter opening range breakout variant with relative volume and time-of-day filters.

Setup	High relative-volume stock breaking a 15-minute opening range near premarket high.	Timeframe	15m execution, 5m confirmation

A link or file upload is transformed into auto-filled fields, generated tags, and strategy information for later graph search.

Generated Output

Auto-filled fields

Title, source type, extracted body text, source tags, generated tags, setup notes, entry rules, exit rules, and risk context.

Example generated tags

strategy:orb · entry:range-break · risk:predefined-stop

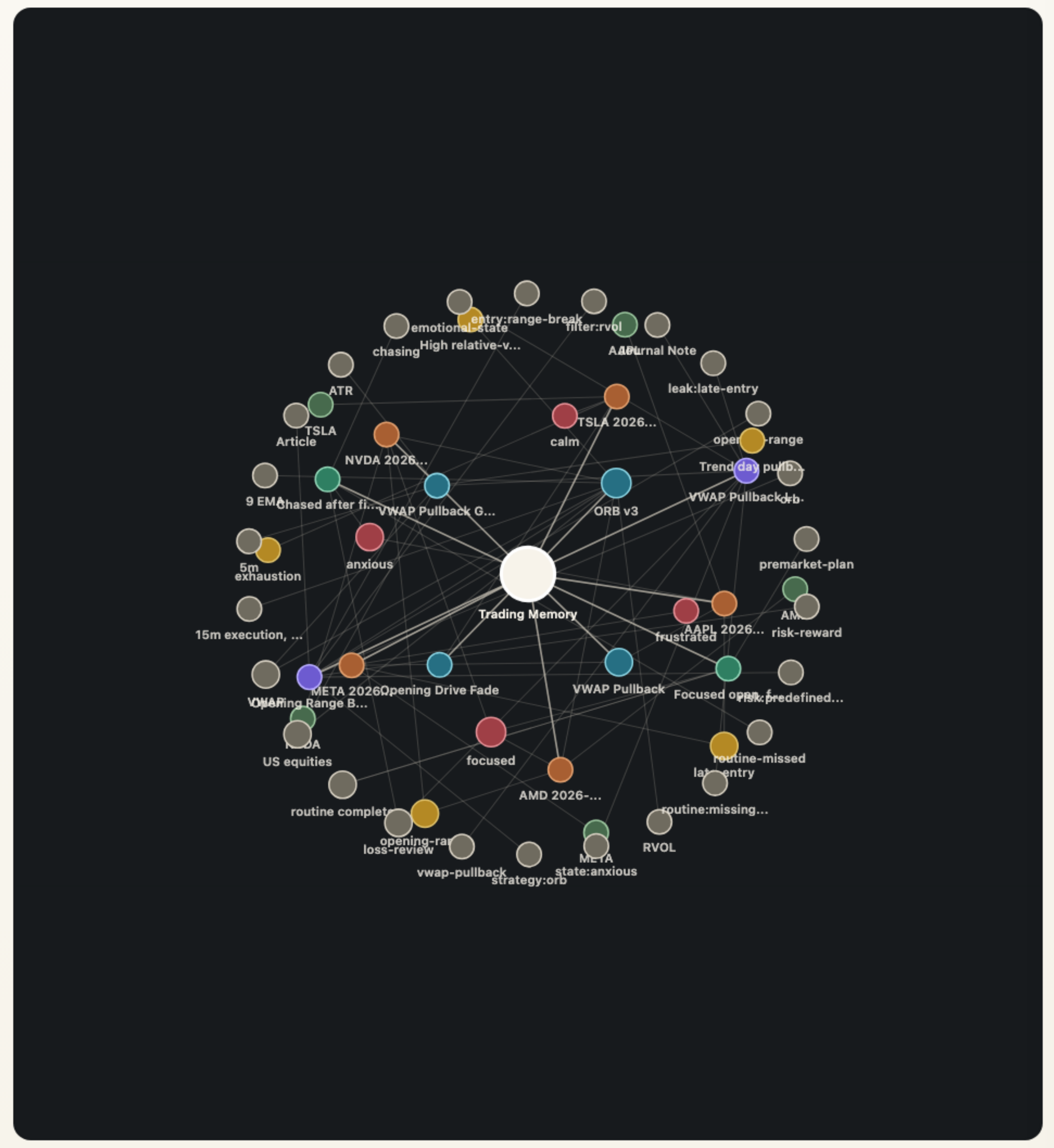
Strategy info created from evidence

Paper-test stricter opening-range breakout rules with relative-volume, time-of-day, and stop-placement filters before any live automation.

Core Memory Graph

Obsidian Map

54 nodes / 68 edges



Node Detail

MEMORY

Trading Memory

5 trades, 2 journal entries, 2 vault sources.

Connections

ORB v3	strategy
Opening Drive Fade	strategy
VWAP Pullback	strategy
NVDA 2026-07-01	trade
AAPL 2026-07-02	trade
TSLA 2026-07-03	trade
AMD 2026-07-04	trade
META 2026-07-05	trade
Focused open, followed plan	journal
Chased after first loss	journal
Opening Range Breakout Playbook	source
VWAP Pullback Loss Review	source

The graph is the product's memory layer: trades, routines, sources, strategies, emotions, symbols, and generated tags become connected nodes.

This is the visual proof that Quant Labs is not only a trade log. It turns scattered evidence into relationships a trader can inspect and query.

System Overview

1 Capture Evidence →
Import links, PDFs, screenshots, scripts, trade notes, and journals.

2 Generate Metadata →
Extract symbols, setups, rules, risks, emotions, entries, exits, and tags.

3 Build Memory Graph →
Connect trades, strategies, routines, sources, journal entries, and tags.

4 Ask Why
Answer review questions using cited evidence instead of isolated trade logs.

Local-first
Browser data works even when the API is offline.

Paper-safe
Generated records can be reviewed before execution.

Evidence-backed
Answers cite trades, notes, screenshots, and sources.

Tech Stack

TypeScript	Next.js	Tailwind	FastAPI	Postgres	Playwright
app logic	workspace	interface	service	storage	screenshots

Evidence + Results

54 graph nodes

68 graph edges

5 closed trades

3 strategies

2 vault sources

Trade Log

Symbol | Side: Long

Date: 07/06/2026

Entry | Exit

Trades

5 closed trades

DATE	SYMBOL	SIDE	STRATEGY	P&
2026-07-01	NVDA	Long	ORB v3	$2
2026-07-02	AAPL	Long	VWAP Pullback	-$
2026-07-03	TSLA	Short	Opening Drive Fade	$2

Seeded trades, strategy labels, journal records, and vault sources generate the current graph and strategy scoreboard.

Why It Matters

Quant Labs makes review reproducible. A trader can move from a vague question to the exact trades, notes, and sources that support the answer.

- TradingView becomes an input, not the whole product.
- The durable asset is the trader's connected memory.
- Explanations stay tied to evidence instead of guesses.

Paper-Only Safety

The MVP is intentionally bounded: it captures, organizes, and explains evidence before any live automation is allowed.

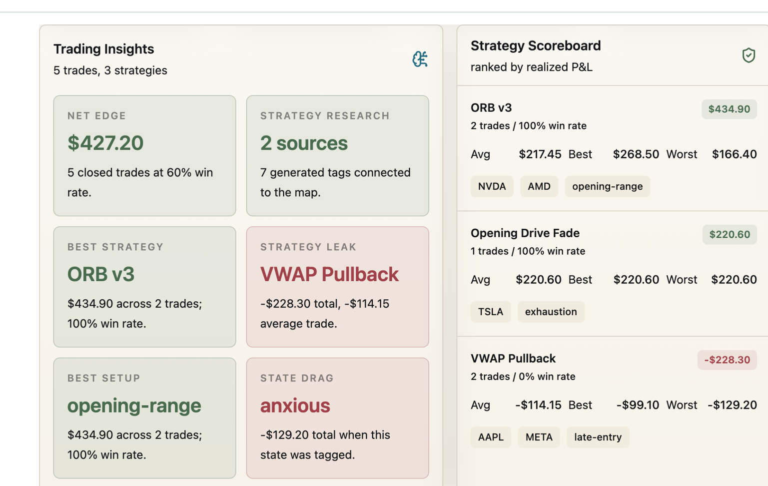
Local-first by default

Browser storage keeps the research workspace usable offline. Future broker and market-data adapters remain outside the core memory model.

Research contribution

The project reframes a trading journal as a graph-backed research engine for reviewing behavior, testing strategy ideas, and finding repeated decision patterns.

Strategy Insight Output



The insights view turns the same evidence into strategy-level review: what worked, what failed, and which setup deserves more paper testing.

Signals produced

working setup · risk cluster · paper-test next

Future Work Roadmap

1 Vector Memory →	2 GraphRAG Chat →	3 Strategy Compiler →	4 Backtest Lab →	5 Provider Mesh →	6 Safety Layer
Embed records.	Cited answers.	Source to rules.	Compare variants.	Data adapters.	Audit gates.

APA Sources

Chauhan, V. (2026). *Quant Labs implementation plan.*

Chauhan, V. (2026). *TradingView adapter thesis.*

Project repository implementation notes.

QR Links

Demo / Repo